

HPS Gaussian-Process Resonance Search

Selected Reconstructed-MC Templates
Standalone Study v6.1

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Abstract

The 2021 Gaussian signal model is compared with normalized reconstructed target-constrained selected-candidate mass distributions from twelve A-prime MC masses between 40 and 260 MeV. The baseline retains the original $\pm 2.25\sigma_m$ fit and training mask, observed 10% spectrum, GP kernel states and coupling normalization. Additional common mask half-widths of 2.4, 2.5 and 2.6 nominal sigma are compared separately. Direct-template observed 90% CL_s limits and local p_0 values at the generated masses are compared with Gaussian fits to the same data. A dense morphing scan is reported separately as exploratory. The MC production is v13 and the observed sample is v16; their target constraint and momentum-sum labels agree, but timing selections differ and full production equivalence is not established. Stored reconstructed masses receive no added smearing or recentering. The inherited coupling coordinate is a conditional comparison, not a validated A-prime exclusion: an unmatched or accidental continuum need not scale with the signal coupling. A further comparison centers the primary fit and training window on an MC-only estimate of the reconstructed core, retaining the nominal width and unchanged MC shape.

1 Observed comparison and scope

The opening comparison retains the original pole-centered prescription. The MC core-centered update appears in Section 12. At the 10 shared generated masses, the median MC-to-Gaussian limit ratio is 0.9876; the range is 0.5332–3.7590. The median absolute relative limit change is 22.05%. The largest absolute relative limit change occurs at 60 MeV (+275.90%). The smallest direct-MC local probability among these tested points occurs at 80 MeV, with $p_0 = 2.477 \times 10^{-2}$ and $Z_0 = 1.964$. This is a sparse-grid local comparison, not a scan-global probability.

The direct comparison uses generated masses from 60 through 240 MeV in 20 MeV steps, inside the original 2021 search. The 40 and 260 MeV samples provide shape and interpolation information outside those direct test points. Full 2015 and full 2016 Gaussian references are retained in the package, but no matching reconstructed MC for those campaigns was supplied. The 2021 response is therefore not substituted into their fits.

The result is conditional on the supplied selected-MC shapes and their finite statistics. It is neither a newly qualified detector-response model nor a coverage calibration. The supplied candidate distributions have broad components and substantial displacement from the generated pole at several masses; they are not established as pure reconstructed resonance responses. An observed difference between the two curves can reflect mass bias, core width, non-Gaussian tails, selection differences and finite-MC fluctuations together.

2 Reconstructed mass distributions

The source histograms retain selected target-constrained masses within 0–400 MeV and record out-of-range counts separately. The pole mass sets the hypothesis; the reconstructed mean is allowed to differ. The comparison does not shift the MC peak onto the pole mass or force its width to the Gaussian parameterization.

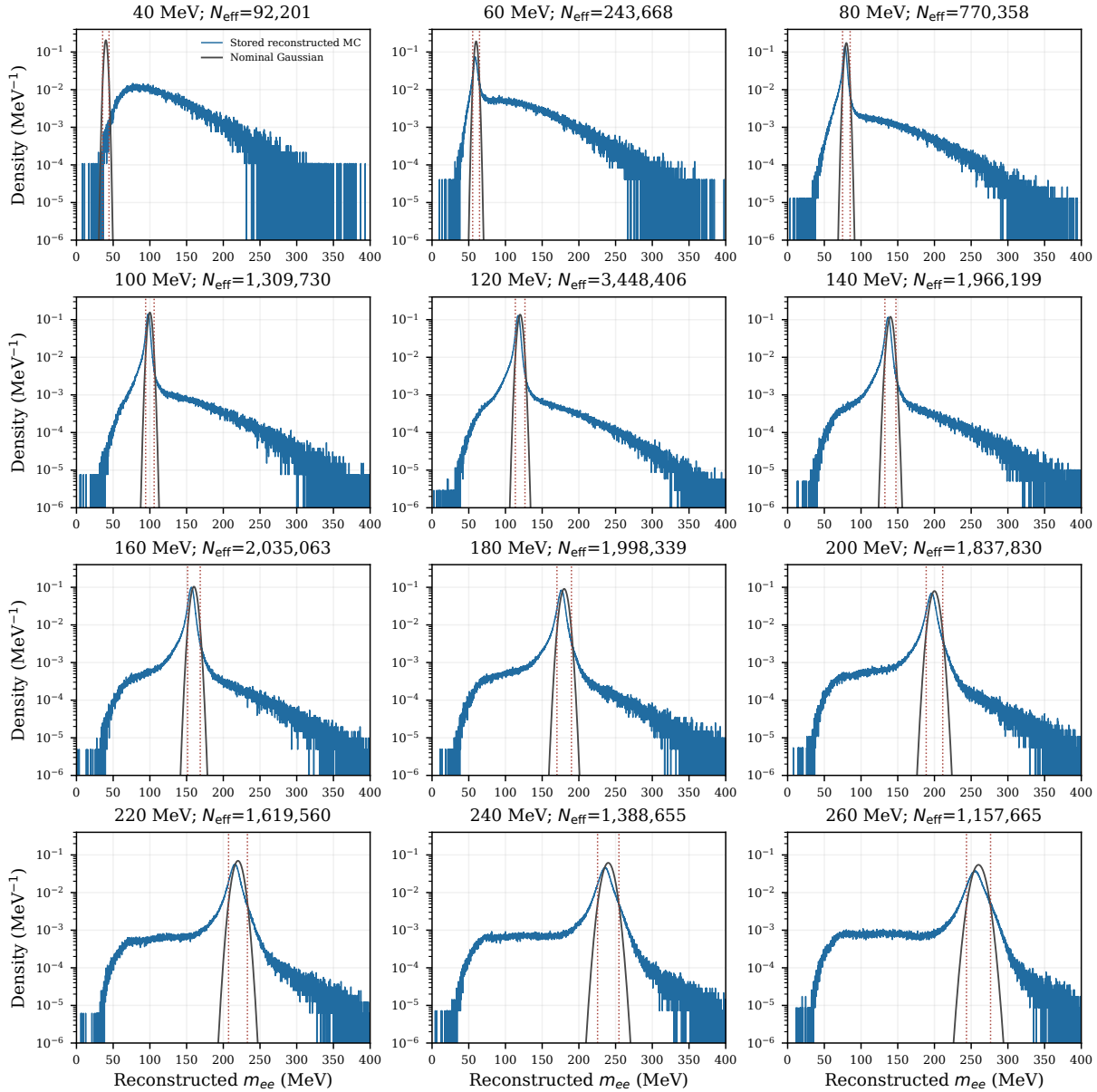


Figure 1: Unit-normalized stored reconstructed MC mass densities and the nominal Gaussian at each generated mass. Densities span the full stored range on a logarithmic axis so broad or displaced populations remain visible. Dotted lines mark the original $\pm 2.25\sigma_m$ boundaries, calculated from the nominal Gaussian resolution. Displayed densities are normalized over their stored histogram range; fitted templates retain their separately declared support normalization. No extra smearing or recentering is applied.

3 Selected MC statistics and shape properties

The source histograms store the weighted bin contents and squared-weight sums. The effective number of weighted entries is

$$N_{\text{eff}} = \frac{(\sum_j w_j)^2}{\sum_j w_j^2}. \quad (1)$$

It describes MC sampling, not data exposure or signal efficiency. Selected rows need not be unique physical events unless the production candidate policy establishes that identity. The source files, latest ROOT tree cycles, branches, weights and extraction counts are recorded in the histogram metadata and manifest.

The stored 0–400 MeV range contains 99.9848–99.9978% of the selected weight across samples; overflow counts are shown separately. All extracted weights are unit weights, so N_{eff} equals the selected count. The raw particle-vector momentum diagnostic falls below 2.8 GeV for 1.35–8.28% of rows. A bounded read finds that stored `psum` and `psum_scalar` satisfy the cut in every inspected row; the raw-vector discrepancy therefore reflects an incompatible convention, not an established selection failure.

m_0 (MeV)	Selected rows	Overflow	Median bias	σ_{68} (MeV)	Core (%)
40	92,201	2	+61.907	40.172	0.53
60	243,668	6	+7.500	40.204	39.18
80	770,358	18	-1.597	7.116	66.92
100	1,309,730	35	-2.411	3.724	72.61
120	3,448,406	123	-2.983	3.898	74.81
140	1,966,199	82	-3.510	4.441	75.64
160	2,035,063	109	-3.966	5.240	75.95
180	1,998,339	129	-4.382	6.311	75.86
200	1,837,830	141	-4.785	7.784	75.17
220	1,619,560	206	-5.250	9.868	73.84
240	1,388,655	180	-5.790	12.854	71.96
260	1,157,665	176	-6.581	17.255	69.90

Table 1: Selected reconstructed MC entries and saved shape summaries. Bias is the reconstructed median displacement, in MeV, from the generated pole. Median and quantile half-width σ_{68} are conditional on the stored 0–400 MeV range; the core probability uses the full selected weight. Overflow denotes selected rows above that range; it remains included in full-selected-yield bookkeeping.

The original 2021 resolution is held fixed when defining every mask:

$$\sigma_m(m) = 0.00184825 - 0.001375m + 0.085875m^2, \quad m, \sigma_m \text{ in GeV}. \quad (2)$$

Thus a narrower or shifted MC core changes the fraction of a unit signal inside the existing fitted bins. It does not silently redefine those bins or the sideband training set. The MC mass range and normalization support are recorded separately so losses at either boundary remain visible.

The word *smear*ed in the production directory is not sufficient to reconstruct the upstream smearing chain. The stored smear-ratio branches and cutflows are audited, while the template itself uses the stored target-constrained mass exactly once. No second factor of 1.25 or undocumented mass transformation is introduced.

4 Observed 2021 limits and local probabilities

The direct MC points use the same observed counts and correlated GP constraint as their Gaussian partners at each mass. The dense morph curve is displayed with a separate line style because its additional mass hypotheses require an interpolation assumption.

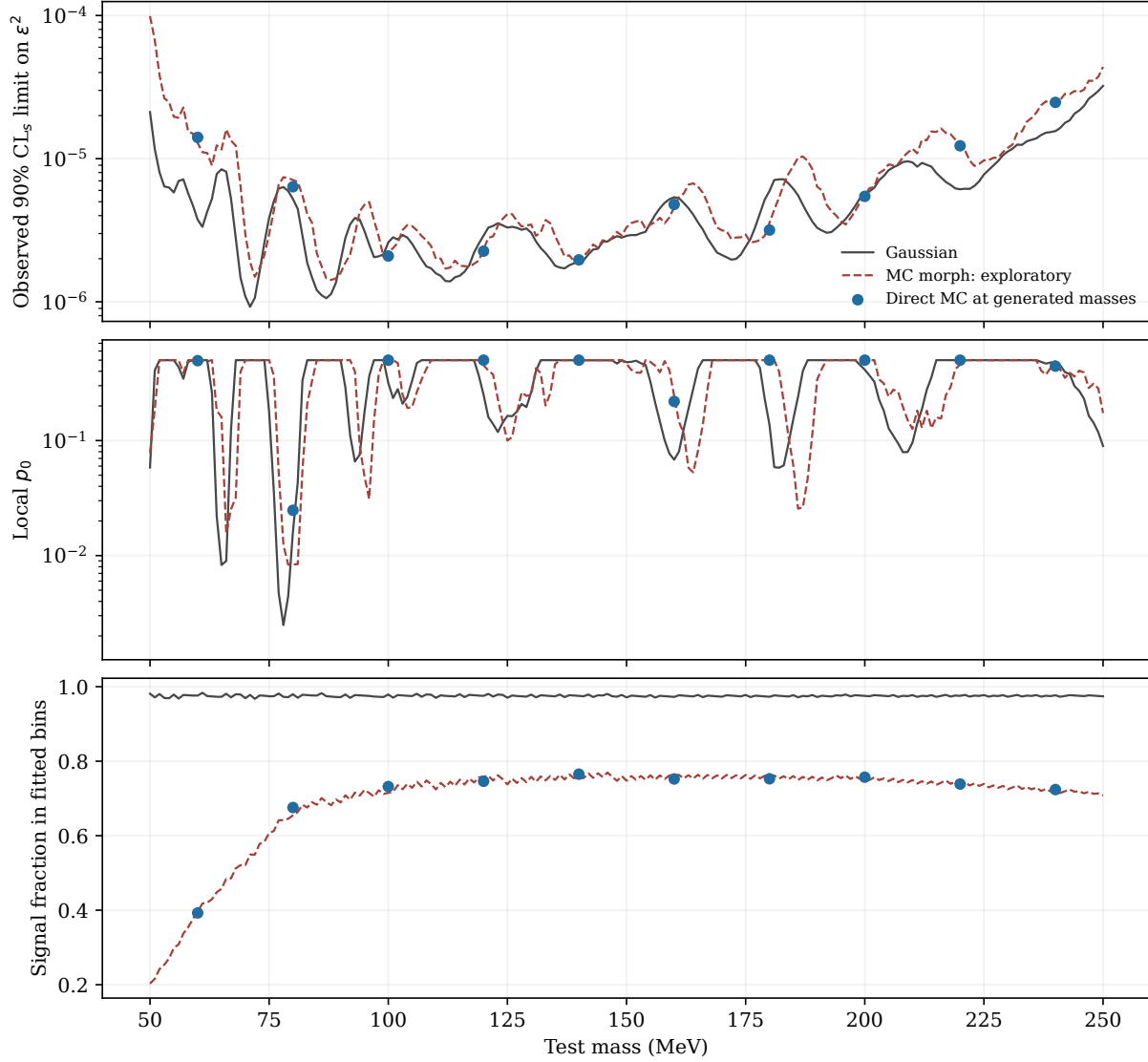


Figure 2: Observed 2021 10% limits, fixed-mass local probabilities and signal fractions in the original fitted bins. Points use the direct reconstructed templates at generated masses. The dashed curve, when present, uses exploratory MC morphing. The coupling limit includes the inherited minimal-visible branching correction once above the dimuon threshold.

5 Paired comparison at generated masses

The direct result is paired with the Gaussian at the same mass. The limit ratio is $R(m) = \varepsilon_{90,MC}^2/\varepsilon_{90,G}^2$, and the signed-root change is $\Delta r = r_{MC} - r_G$. Points at different masses share observations and GP training bins.

m_0	$\varepsilon_{90,G}^2$	$\varepsilon_{90,MC}^2$	Ratio	$p_{0,G}$	$p_{0,MC}$	Δr
60	3.747×10^{-6}	1.408×10^{-5}	3.7590	5.000×10^{-1}	4.950×10^{-1}	+0.825
80	5.221×10^{-6}	6.362×10^{-6}	1.2184	1.628×10^{-2}	2.477×10^{-2}	-0.174
100	2.599×10^{-6}	2.089×10^{-6}	0.8039	3.173×10^{-1}	5.000×10^{-1}	-0.969
120	2.905×10^{-6}	2.258×10^{-6}	0.7773	2.500×10^{-1}	5.000×10^{-1}	-0.969
140	1.968×10^{-6}	1.964×10^{-6}	0.9975	5.000×10^{-1}	5.000×10^{-1}	-0.356
160	5.376×10^{-6}	4.787×10^{-6}	0.8904	6.817×10^{-2}	2.188×10^{-1}	-0.713
180	5.946×10^{-6}	3.170×10^{-6}	0.5332	1.371×10^{-1}	5.000×10^{-1}	-2.194
200	5.577×10^{-6}	5.453×10^{-6}	0.9777	4.128×10^{-1}	5.000×10^{-1}	-0.684
220	6.114×10^{-6}	1.230×10^{-5}	2.0113	5.000×10^{-1}	5.000×10^{-1}	+1.117
240	1.558×10^{-5}	2.472×10^{-5}	1.5865	4.865×10^{-1}	4.436×10^{-1}	+0.108

Table 2: Direct-template comparison at generated masses. All probabilities are pointwise and use their stated signal template.

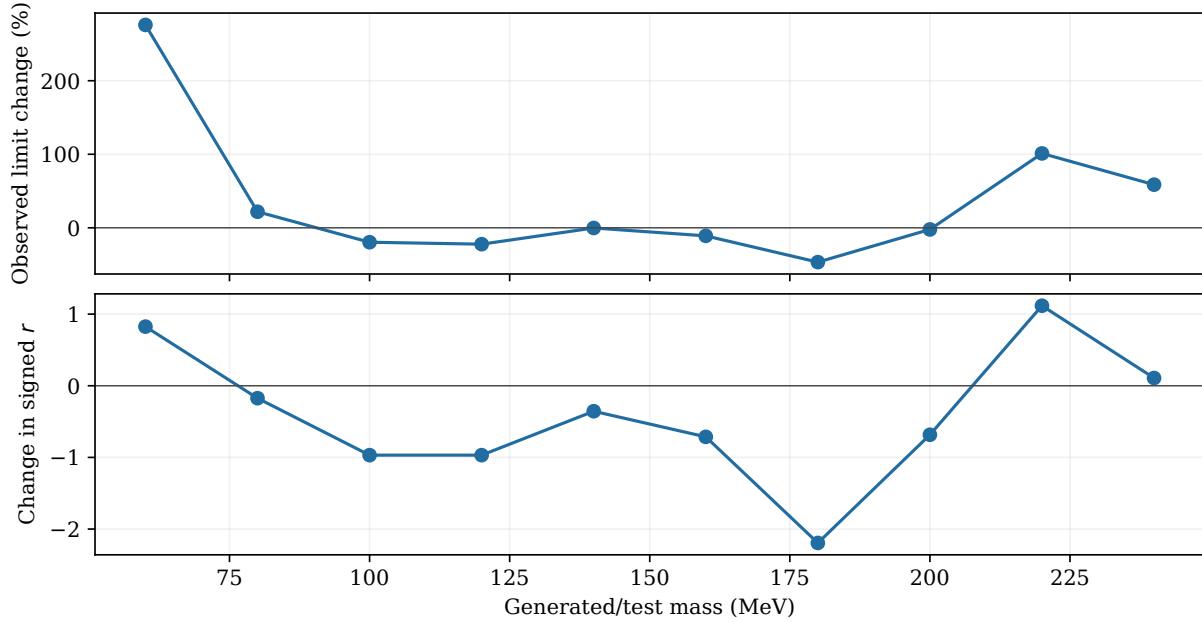


Figure 3: Observed fractional limit changes and signed-root differences for direct MC templates. These are paired descriptions of the same observed spectra. They are not independent residuals with a chi-squared sampling law.

6 Agreement summaries and interpolation boundary

Descriptive quantity	Direct MC versus Gaussian
Common generated masses	10
Median ratio	0.98760
16th–84th percentile ratio	0.78900–1.82440
Minimum / maximum ratio	0.53322 / 3.75896
Median absolute relative change	22.055%
RMS log ratio	0.54902
Within 1 / 5 / 10%	10.0% / 20.0% / 20.0%
Maximum $ \Delta r $	2.19398

Table 3: Descriptive agreement on the common direct-MC grid. Quantile ranges and percentages of agreeing masses are summaries of the curves, not confidence intervals or agreement probabilities.

A statistical probability for disagreement would require a specified joint sampling calculation that preserves the common data, overlapping sidebands and template estimation. An independent-point chi-squared test would omit those dependencies. The present comparison therefore reports sizes and locations of differences without assigning such a probability.

The reconstructed MC is available at discrete generated masses. A dense scan necessarily interpolates shape information. Holding out an interior generated mass and predicting its shape from its neighbors tests that interpolation against an available simulation sample. It does not test the underlying detector simulation against data, and it cannot establish sampling coverage of an upper limit.

The declared morph linearly mixes the neighboring CDFs in standardized residual $z = (m_{\text{rec}} - m_0)/\sigma_{\text{nominal}}(m_0)$, evaluating each input distribution at its own pole and nominal resolution. It uses no extrapolation beyond the supplied 40–260 MeV grid. Its dense observed scan covers the original 50–250 MeV search at 1 MeV spacing.

8 of 10 held-out shape checks pass the declared $D_{\text{CDF}} \leq 0.03$ and $|\Delta F_{\text{core}}| \leq 0.02$ criteria. The largest conditional-support CDF distance is 0.1395. The held-out limit ratios to direct MC range from 0.9142 to 1.3029. The dense curve remains exploratory regardless of this outcome.

The direct generated-mass results remain the primary comparison. Any minimum found on the denser exploratory grid depends on both the morphing prescription and the enlarged mass scan. Neither its local probability nor the smallest value selected across models is a global discovery probability.

6.1 Held-out shapes and finite-MC precision

m_0	D_{CDF}	ΔF_{core}	UL ratio	Δr	Shape check
60	0.1168	-0.0545	0.9142	+0.3724	fail
80	0.1395	-0.1090	1.3029	+0.0490	fail
100	0.0271	-0.0166	1.0540	+0.0364	pass
120	0.0118	-0.0062	1.0230	+0.0116	pass
140	0.0091	-0.0025	1.0117	+0.0037	pass
160	0.0057	-0.0016	1.0143	+0.0103	pass
180	0.0037	-0.0028	1.0165	+0.0124	pass
200	0.0033	-0.0030	1.0056	-0.0154	pass
220	0.0027	-0.0026	1.0177	+0.0289	pass
240	0.0052	-0.0006	1.0024	+0.0168	pass

Table 4: Leave-one-mass-out checks. CDF distances and core fractions use the conditional analysis-support distributions. Limit ratios and signed-root changes compare the held-out morph with direct MC at the same hypothesis. The shape thresholds are diagnostic criteria.

For each direct mass, 32 independent Poisson resamples of the stored unit-weight MC histogram bins are fitted against the fixed Gaussian reference. This conditional calculation quantifies template sampling precision under that bin-count model. It does not include detector systematics, selection differences, correlations between candidates or uncertainty in the observed-data background. No new observed-data experiments are generated.

m_0	Resamples	R_{16}	R_{50}	R_{84}	UL spread (%)
60	32	3.7284	3.7609	3.7925	1.063
80	32	1.2141	1.2173	1.2213	0.349
100	32	0.8016	0.8037	0.8061	0.302
120	32	0.7767	0.7772	0.7780	0.096
140	32	0.9960	0.9978	0.9987	0.125
160	32	0.8887	0.8908	0.8920	0.212
180	32	0.5320	0.5332	0.5350	0.243
200	32	0.9750	0.9773	0.9798	0.268
220	32	2.0067	2.0148	2.0185	0.312
240	32	1.5794	1.5852	1.5907	0.326

Table 5: Conditional finite-MC precision. R_{16}, R_{50}, R_{84} are quantiles of the MC-to-Gaussian upper-limit ratio across 32 MC-bin resamples. The last column is the relative standard deviation of the resampled limit around its central direct-template scale, in percent.

Thirty-two resamples resolve only a modest estimate of the central spread. The intervals are neither an observed-limit confidence band nor a significance for the discrepancy between templates.

7 Likelihood and normalization

The baseline uses the original primary window. The GP trains only on bins outside $\pm 2.25\sigma_m$, using archived kernel coordinates. The window diagnostics replace 2.25 by 2.4, 2.5 or 2.6 for both the fit and training mask, recomputing the prediction with the same kernel states. Within a mass hypothesis, both signal models share its predicted count mean b , covariance C and observed counts n . With $LL^T = C$,

$$a = \frac{\varepsilon^2}{10^{-8}}, \quad T_i = K(m) 10^{-8} w_i, \quad (3)$$

$$\lambda_i = b_i + (L\theta)_i + aT_i, \quad (4)$$

$$\ell(a, \theta) = \sum_i \left[\lambda_i - n_i + n_i \log \frac{n_i}{\lambda_i} \right] + \frac{1}{2} \theta^T \theta. \quad (5)$$

The zero-count contribution is λ_i . The correlated background deformation is profiled at every tested amplitude. The bin probabilities w_i are normalized over the declared full support before selecting fitted bins; their restriction to the primary window is not renormalized.

For MC the full fitting support is 36–299.75 MeV. Its retained fraction f_{support} is recorded separately. The ledger’s `full_yield_90` is the support-normalized yield; the additional `full_selected_yield_90` divides it by f_{support} . That bookkeeping conversion does not establish a calibrated coupling normalization outside the analysis support.

Here $K(m)$ is the unchanged prompt-density conversion. The solver’s `A90` column is a_{90} , while the support-normalized event yield is $N_{90} = a_{90} K(m) 10^{-8} = K(m) \varepsilon_{90}^2$. Above $2m_\mu$, the inherited minimal-visible conversion multiplies the electron-only coupling limit by $1 + \sqrt{1 - 4r_\mu}(1 + 2r_\mu)$, with $r_\mu = (m_\mu/m)^2$. This correction does not change the visible yield or local probability. Density continues to use the nominal $m \pm 1.64\sigma_m$ native-bin interval.

The observed limit solves the bounded, piecewise-asymptotic $CL_s(a_{90}) = 0.10$, using explicitly profiled observed and background-Asimov likelihood ratios [2]. The free diagnostic amplitude may be negative while physical Poisson expectations are maintained. Writing its signed likelihood root as r ,

$$Z_0 = \max(r, 0), \quad p_0 = \Phi(-Z_0). \quad (6)$$

Deficits therefore have the inherited reference value $p_0 = 0.5$. This is a local asymptotic reference, not the probability of the most significant result anywhere in a scan.

The Gaussian and MC comparisons use the same converged solver. The historical v5.9 audit found a stopping discrepancy in an older optimizer; the present paired comparison does not mix that older table with newly computed MC fits. Numerical convergence and finite-MC statistical qualification are separate questions.

8 Selection provenance and interpretation

The supplied MC directory is `/sdf/data/hps/physics2021/preselection/v13/ap_signal_prompt_smeared`. The observed input derives from `/sdf/data/hps/physics2021/preselection/v16/data_10pc_prompt_TC_psum2p8`. The MC uses the stored `vertex./vertex.invM_` branch and the target-constrained vertex category. Production labels agree on the target constraint and $p_{\text{sum}} > 2.8$ GeV, but the inspected MC timing windows include 5.7 and 9.2 ns where the observed sample uses 5.1 and 7.8 ns. Full production equivalence is not established. No new selection is silently imposed to make the two samples appear matched.

A bounded inspection finds both stored daughter truth-link flags false for every inspected row. Those flags cannot identify matched signal daughters, and their values alone do not prove that the reconstructed candidates lack signal daughters. No additional truth selection is defensible from them. The fit therefore substitutes the supplied all-selected candidate distribution conditionally.

The stored track hit counters are three-dimensional; cutflow labels refer to two-dimensional-hit requirements. Those labels are not a license to apply their thresholds directly to the stored hit counters. The local and remote selection audits preserve the evidence and the unresolved production details. ROOT files can contain older tree cycles; only the latest `preselection` cycle contributes to extraction.

Input files and configurations are identified by saved paths, sizes and SHA-256 hashes. Extraction uses bounded chunks and retains bin sums of weights and squared weights. All results, scripts, reconstructed-mass histograms and report sources are mirrored to `/sdf/home/e/epeets/src/hps_gpr_v6p1_mc_signal_templates_20260922`. The original observed result and source ROOT files are preserved.

This study measures how the selected reconstructed 2021 MC changes the conditional observed inference under the archived background model and the explicitly declared fit and training mask. It does not isolate a pure tail uncertainty or establish that either signal template is the correct detector response. Selection differences, finite MC and interpolation remain explicit qualifications. No MC-template claim is made for the 2015 or 2016 detector configurations.

Approximately 24–61% of the direct-mass selected signal distributions lie outside the primary blind window. Signal in those training bins can be absorbed by a background-only GP. This observed template replacement keeps the sideband prediction fixed; it does not model signal-contamination feedback, extraction recovery or coverage for these broad distributions.

References

- [1] E. Peets, M. Gignac and E. Hsu, *HPS Gaussian-Process Resonance Search: Analysis Note v5.0.5*, 16 September 2026, internal review draft and archived source package.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, “Asymptotic formulae for likelihood-based tests of new physics,” *Eur. Phys. J. C* **71**, 1554 (2011), arXiv:1007.1727.

9 Common fit and training windows of 2.4, 2.5 and 2.6 sigma

The original $\pm 2.25\sigma_m$ prescription remains the baseline. Three additional calculations use common fit and training-mask half-widths $h = 2.4\sigma_m$, $2.5\sigma_m$ and $2.6\sigma_m$. There is no separate outer guard window: the same half-width determines the fitted bins and the excluded GP training bins. Each changed mask recomputes the GP prediction and correlated covariance from its own observed sidebands, with the archived kernel coordinates held fixed.

Gaussian and MC templates at a given width share those counts and GP constraints. A comparison between widths therefore changes both the observed fit region and the background prediction; it cannot isolate signal-tail acceptance alone. The nominal resolution, coupling conversion and full-support template normalization are retained. Masks select bins by their native-bin centers. Nearby half-widths can therefore select exactly the same fit and training bins at a particular mass and yield coincident results; the width response is not continuous. No width is selected for its observed limit or local probability.

h/σ_m	Median ratio	Ratio range	Med. $ R - 1 $ (%)	Max. $ \Delta r $	LOO pass	MC spread (%)
2.25	0.9876	0.5332–3.7590	22.05	2.194	8/10	1.06
2.4	1.0038	0.5413–3.7386	22.92	2.164	8/10	1.11
2.5	1.0127	0.5480–3.9413	28.90	2.151	8/10	1.13
2.6	1.0127	0.5491–3.9413	29.53	2.170	8/10	1.13

Table 6: Direct-MC versus Gaussian comparisons at ten generated masses, reported separately for each common half-width. The LOO column counts held-out shape checks satisfying the declared criteria. MC spread is the largest relative upper-limit standard deviation among the ten masses under 32 independent-bin Poisson resamples. These descriptions are correlated across both mass and window.

The median absolute MC-to-Gaussian limit change is 22.92%, 28.90% and 29.53% for half-widths 2.4, 2.5 and 2.6 sigma, compared with 22.05% at baseline. The smallest direct-template local probability remains at 80 MeV: $p_0 = 0.02477$ for baseline and 2.4 sigma, and $p_0 = 0.009903$ for 2.5 and 2.6 sigma. These are conditional local results, not a basis for choosing the width.

The following pages keep the three new prescriptions separate. Their upper panels show the full observed curves; the lower panels and tables compare direct templates only at their generated masses. Dashed curves use the same exploratory CDF morph as the baseline, including its low-mass qualification. Local probabilities retain $p_0 = 0.5$ for deficits and are not adjusted for inspecting multiple masses, models or window widths.

The finite-MC and leave-one-out diagnostics are recomputed for each width and retained in the portable ledgers. Wider masks do not resolve the unverified candidate association, production-selection mismatch or signal leakage into the sidebands. The inherited coupling values remain conditional template comparisons rather than validated A-prime exclusions.

9.1 Half-width $2.4\sigma_m$

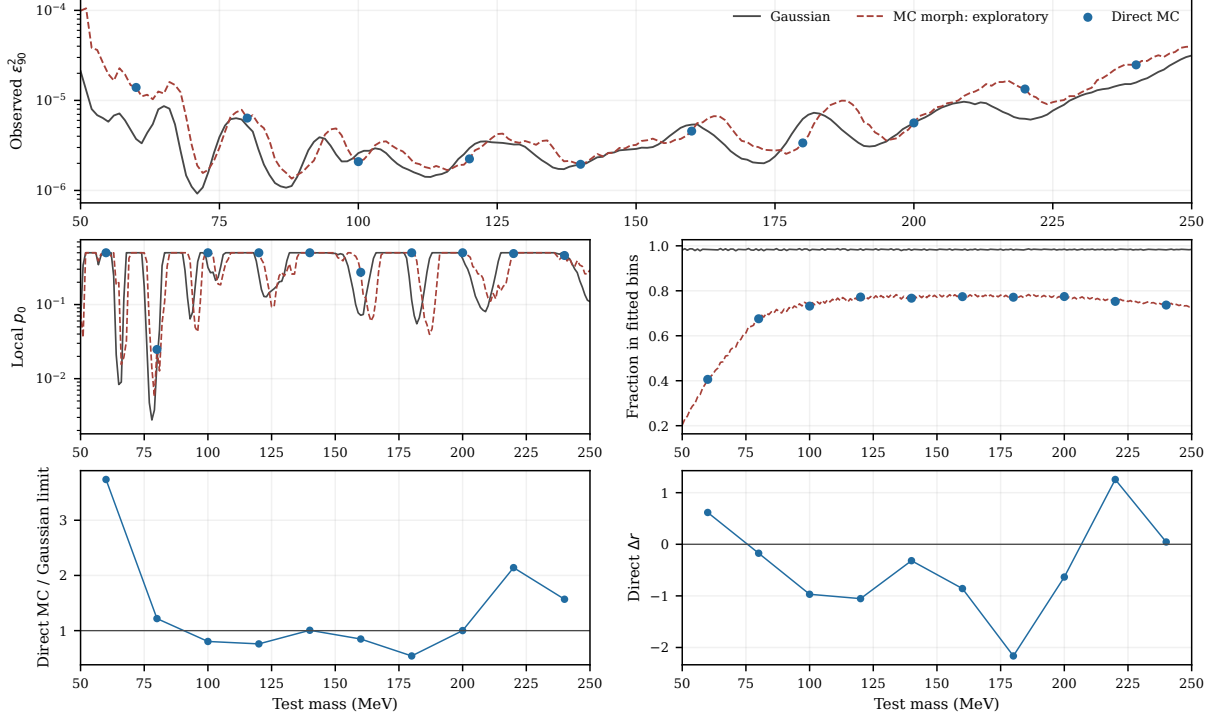


Figure 4: 2021 10% observed results with common fit and GP training-mask half-width $2.4\sigma_m$. The Gaussian and MC predictions share the background constraint within this page. The lower panels show direct-MC/Gaussian limit ratios and signed-root changes at generated masses; dashed dense curves are exploratory.

m_0	$\varepsilon_{90,G}^2$	$\varepsilon_{90,MC}^2$	Ratio	$p_{0,G}$	$p_{0,MC}$	Δr
60	3.73×10^{-6}	1.39×10^{-5}	3.739	0.5000	0.5000	+0.616
80	5.22×10^{-6}	6.36×10^{-6}	1.218	0.0163	0.0248	-0.174
100	2.60×10^{-6}	2.09×10^{-6}	0.804	0.3173	0.5000	-0.969
120	2.95×10^{-6}	2.24×10^{-6}	0.760	0.2465	0.5000	-1.053
140	1.94×10^{-6}	1.95×10^{-6}	1.007	0.5000	0.5000	-0.318
160	5.38×10^{-6}	4.56×10^{-6}	0.848	0.0716	0.2724	-0.858
180	6.23×10^{-6}	3.37×10^{-6}	0.541	0.1148	0.5000	-2.164
200	5.62×10^{-6}	5.62×10^{-6}	1.001	0.4175	0.5000	-0.635
220	6.25×10^{-6}	1.34×10^{-5}	2.141	0.5000	0.4879	+1.256
240	1.58×10^{-5}	2.48×10^{-5}	1.569	0.4771	0.4595	+0.044

Table 7: Direct-template comparison for half-width $2.4\sigma_m$. Coupling limits and local probabilities retain the same conventions as the baseline table.

9.2 Half-width $2.5\sigma_m$

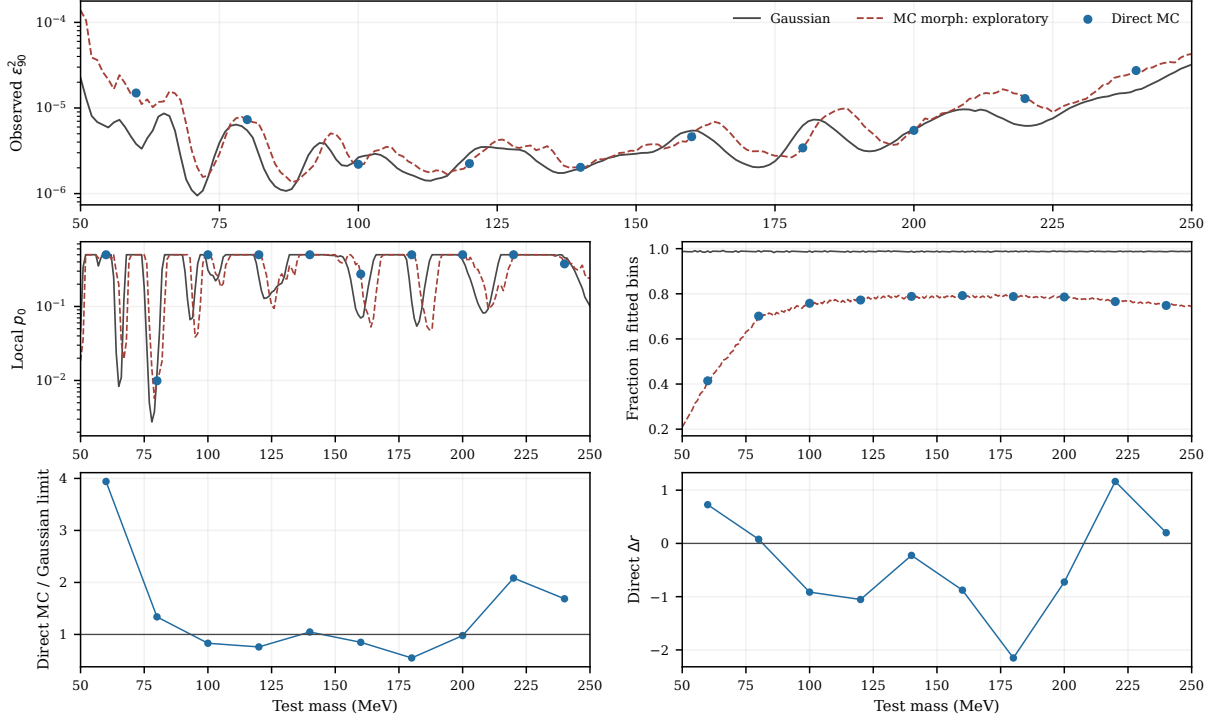


Figure 5: 2021 10% observed results with common fit and GP training-mask half-width $2.5\sigma_m$. The Gaussian and MC predictions share the background constraint within this page. The lower panels show direct-MC/Gaussian limit ratios and signed-root changes at generated masses; dashed dense curves are exploratory.

m_0	$\varepsilon_{90,G}^2$	$\varepsilon_{90,MC}^2$	Ratio	$p_{0,G}$	$p_{0,MC}$	Δr
60	3.80×10^{-6}	1.50×10^{-5}	3.941	0.5000	0.5000	+0.727
80	5.47×10^{-6}	7.31×10^{-6}	1.338	0.0121	0.0099	+0.078
100	2.65×10^{-6}	2.20×10^{-6}	0.831	0.3101	0.5000	-0.915
120	2.95×10^{-6}	2.24×10^{-6}	0.760	0.2465	0.5000	-1.053
140	1.94×10^{-6}	2.03×10^{-6}	1.047	0.5000	0.5000	-0.225
160	5.44×10^{-6}	4.62×10^{-6}	0.849	0.0698	0.2743	-0.878
180	6.24×10^{-6}	3.42×10^{-6}	0.548	0.1170	0.5000	-2.151
200	5.60×10^{-6}	5.47×10^{-6}	0.978	0.4234	0.5000	-0.725
220	6.18×10^{-6}	1.29×10^{-5}	2.084	0.5000	0.5000	+1.163
240	1.62×10^{-5}	2.74×10^{-5}	1.686	0.4545	0.3763	+0.201

Table 8: Direct-template comparison for half-width $2.5\sigma_m$. Coupling limits and local probabilities retain the same conventions as the baseline table.

9.3 Half-width $2.6\sigma_m$

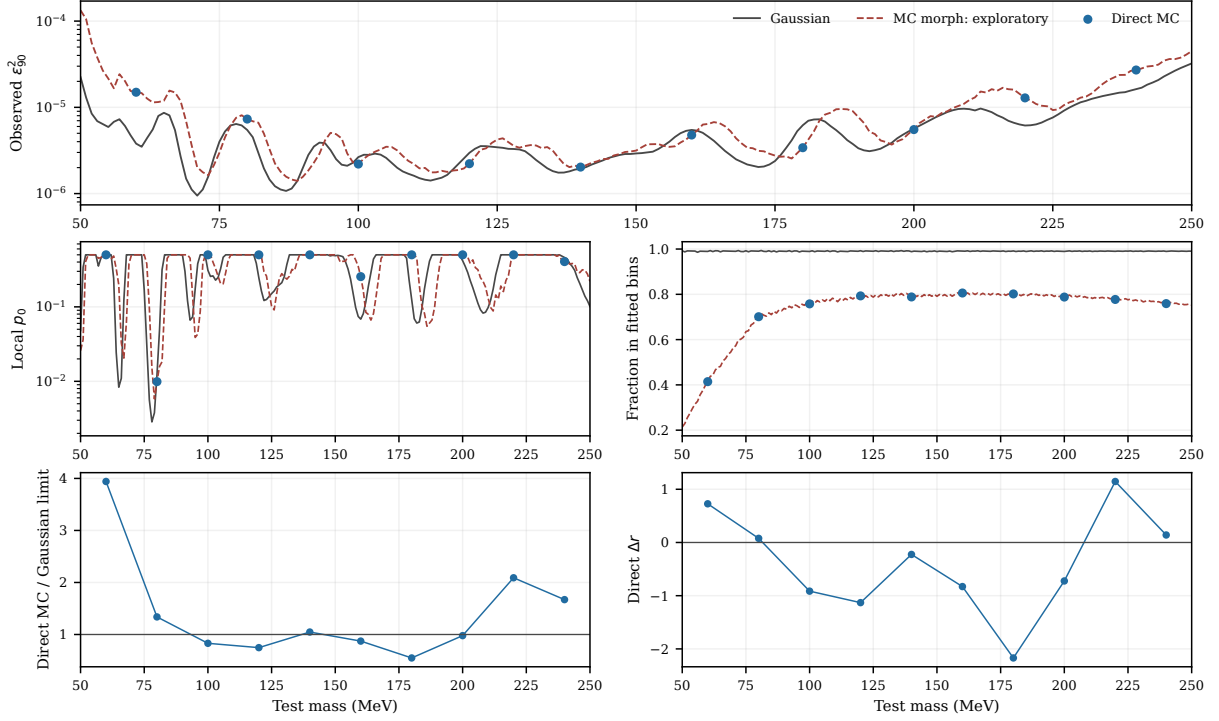


Figure 6: 2021 10% observed results with common fit and GP training-mask half-width $2.6\sigma_m$. The Gaussian and MC predictions share the background constraint within this page. The lower panels show direct-MC/Gaussian limit ratios and signed-root changes at generated masses; dashed dense curves are exploratory.

m_0	$\varepsilon_{90,G}^2$	$\varepsilon_{90,MC}^2$	Ratio	$p_{0,G}$	$p_{0,MC}$	Δr
60	3.80×10^{-6}	1.50×10^{-5}	3.941	0.5000	0.5000	+0.727
80	5.47×10^{-6}	7.31×10^{-6}	1.338	0.0121	0.0099	+0.078
100	2.65×10^{-6}	2.20×10^{-6}	0.831	0.3101	0.5000	-0.915
120	2.97×10^{-6}	2.22×10^{-6}	0.747	0.2454	0.5000	-1.131
140	1.94×10^{-6}	2.03×10^{-6}	1.047	0.5000	0.5000	-0.225
160	5.47×10^{-6}	4.78×10^{-6}	0.873	0.0684	0.2549	-0.829
180	6.20×10^{-6}	3.40×10^{-6}	0.549	0.1234	0.5000	-2.170
200	5.65×10^{-6}	5.52×10^{-6}	0.978	0.4173	0.5000	-0.723
220	6.16×10^{-6}	1.29×10^{-5}	2.090	0.5000	0.5000	+1.146
240	1.62×10^{-5}	2.71×10^{-5}	1.670	0.4606	0.4053	+0.141

Table 9: Direct-template comparison for half-width $2.6\sigma_m$. Coupling limits and local probabilities retain the same conventions as the baseline table.

10 Morphed midpoint templates: full reconstructed range

The eleven midpoint hypotheses use equal CDF mixtures of their two neighboring generated samples, transported through the nominal-resolution standardized coordinate. The plots use saved numerical morphs, not a fit to observed data. The bracketing curves show the transported inputs at the midpoint hypothesis. Each displayed curve is normalized over 0–400 MeV; fitted probabilities retain the separately recorded analysis support normalization.

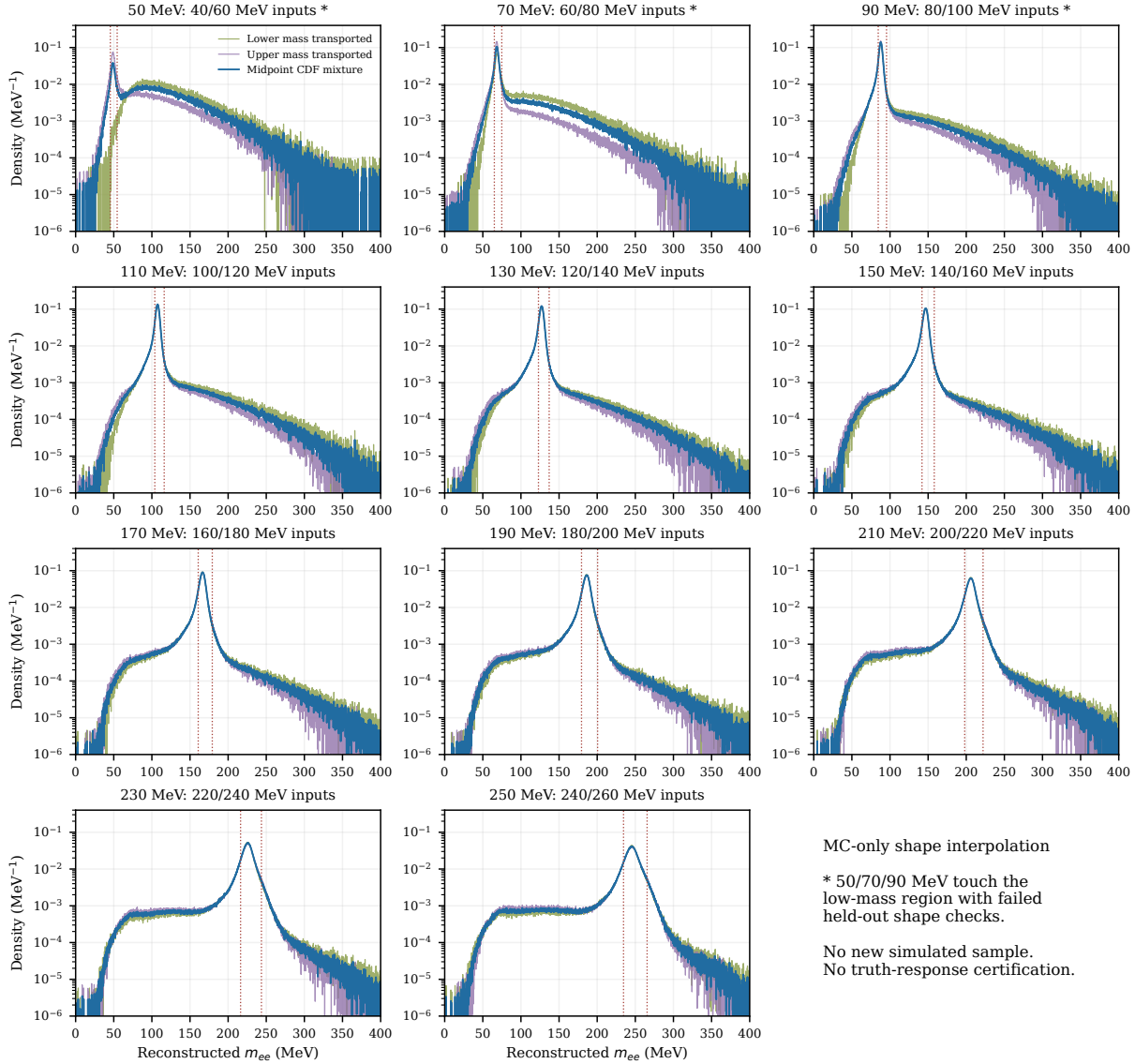


Figure 7: Full-range reconstructed mass densities for midpoint morphs at 50, 70, ..., 250 MeV, with transported neighboring MC shapes. The logarithmic scale exposes broad and displaced populations. Dotted lines mark the baseline $\pm 2.25\sigma_m$ interval for reference; the morph itself is independent of the fit-window width. Asterisks flag the 50/70/90 MeV hypotheses that touch the low-mass region with failed held-out shape checks.

11 Morphed midpoint templates: core detail

The same saved distributions are displayed within seven nominal sigma of each midpoint, on a linear density scale. Zooming the horizontal axis does not renormalize the curves or remove their broad components from the fitted signal model. The interpolation supplies no additional truth matching, production-equivalence validation or detector-response certification.

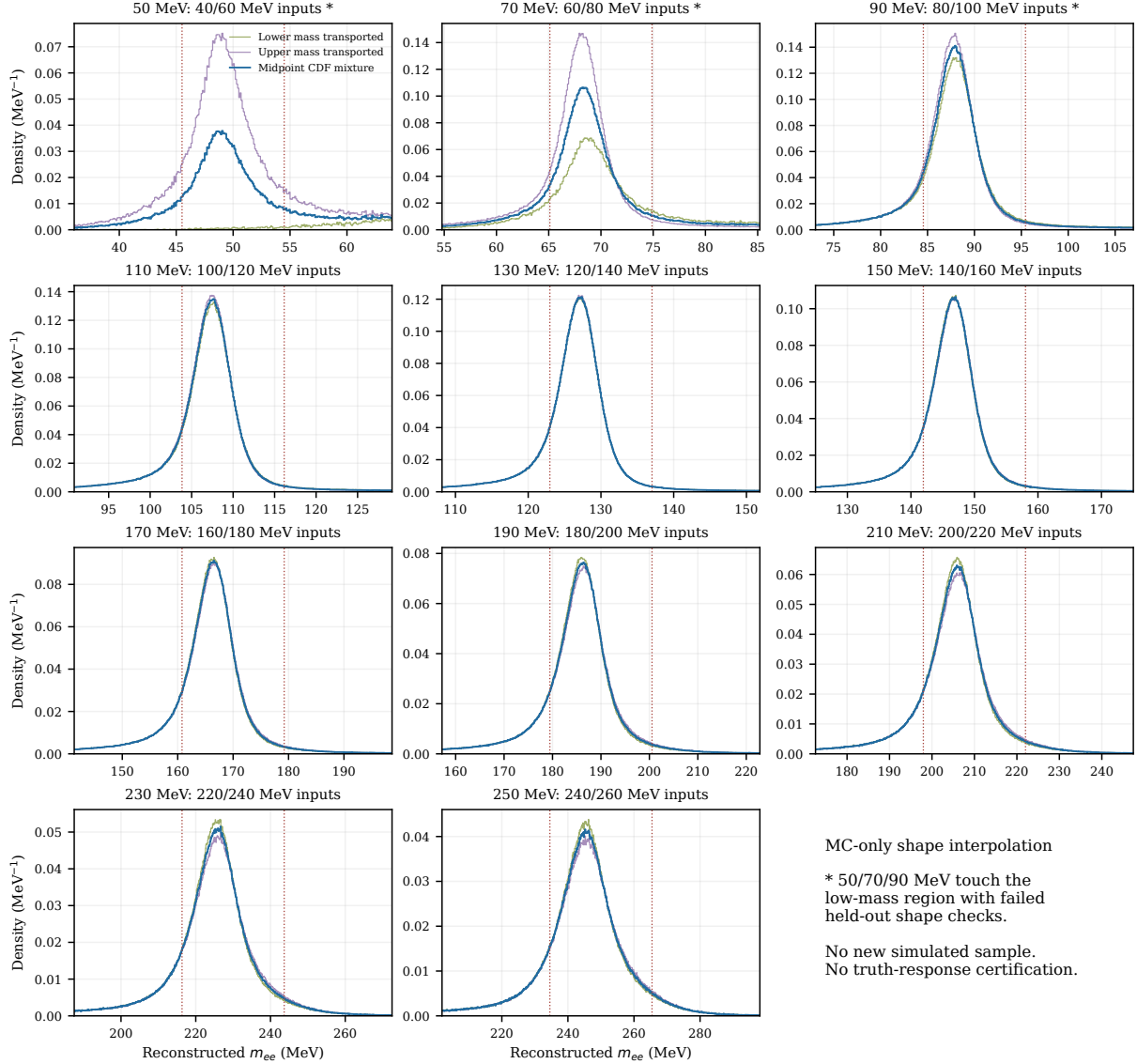


Figure 8: Core detail of the same midpoint morphs and transported neighboring shapes. The plot scale is set independently in each panel. Dotted boundaries retain the original baseline half-width. These MC-only interpolations remain exploratory, particularly the marked low-mass hypotheses; they are not new simulated samples.

12 MC core centers and shifted primary windows

This update, dated 23 September 2026, centers the observed fit and the excluded GP training interval on the reconstructed MC core, rather than the generated pole. The reconstructed template itself is unchanged. For a generated hypothesis m_0 and an MC-only core location $c(m_0)$,

$$I(m_0) = [c(m_0) - 2.25\sigma_{\text{nominal}}(m_0), c(m_0) + 2.25\sigma_{\text{nominal}}(m_0)]. \quad (7)$$

The half-width retains the original nominal resolution. The fitted core width below does not replace it. No additional guard is used.

The center is determined without observed counts. A Gaussian smoothing of width $0.2\sigma_{\text{nominal}}$ locates the largest MC peak within $m_0 \pm 3\sigma_{\text{nominal}}$. A bin-integrated Gaussian plus a nonnegative affine pedestal is then fitted by Poisson deviance to the unsmoothed template within $\pm 1.5\sigma_{\text{nominal}}$ of that mode. The Gaussian mean defines c ; the pedestal limits the influence of a local broad component. The mean may vary by one nominal sigma around the mode, and the fitted width by 0.25–2.5 nominal sigma. All 201 fits converge without these bounds being active. This is an operational core definition, not proof of a pure resonance response.

m_0	$c(m_0)$	$c - m_0$	σ_{core}	MC spread	Max. $ \Delta c $	Max. UL change (%)
60	58.792	-1.208	1.550	0.034	0.049	0.00
80	78.103	-1.897	1.637	0.007	0.003	0.00
100	97.688	-2.312	1.853	0.006	0.027	0.00
120	117.319	-2.681	2.112	0.003	0.039	0.00
140	136.956	-3.044	2.392	0.006	0.052	0.29
160	156.658	-3.342	2.713	0.007	0.081	0.00
180	176.393	-3.607	3.120	0.008	0.096	0.00
200	196.182	-3.818	3.577	0.011	0.130	0.00
220	215.941	-4.059	4.233	0.012	0.162	4.01
240	235.617	-4.383	4.981	0.022	0.132	3.75

Table 10: Native-MC core centers and stability. All mass quantities are in MeV. MC spread is the standard deviation of the estimated center under 32 independent-bin Poisson resamples. The last two columns vary the core-fit half-width from its default 1.5 nominal sigma to 1.25 and 2.0, and refit the observed MC limit using each alternative center.

The maximum center-definition displacement is 0.162 MeV and the largest corresponding limit change is 4.01%. Native-bin masks are discrete; many smaller shifts select exactly the same bins. Bin resampling quantifies conditional center precision only; it does not establish candidate independence or cover production systematics.

The direct-template centers shift by 1.21–4.38 MeV below their generated poles. The median absolute MC/Gaussian limit difference in the matched centered windows is 17.83%, compared with 22.05% for the original paired prescription. The native new/old MC limit ratios span 0.822–1.201. The smallest native MC local probability is at $m_0 = 80$ MeV, $c = 78.103$ MeV: $p_0 = 0.00787$ ($Z_0 = 2.415$), compared with $p_0 = 0.02477$ before shifting the window.

12.1 Observed scans after centering

The GP mean and covariance are recomputed from sidebands outside each shifted interval. Archived kernel coordinates remain anchored at m_0 . The inherited density conversion, nominal resolution and branching conversion also retain m_0 , isolating the center change from a change of coupling convention. Both the MC and its new Gaussian benchmark use the same shifted window and GP constraint. The benchmark Gaussian is centered at $c(m_0)$ with width $\sigma_{\text{nominal}}(m_0)$; it is not the historical Gaussian fit evaluated at mass c with new kernel or density settings. All curves are plotted against generated hypothesis m_0 , not reconstructed center c .

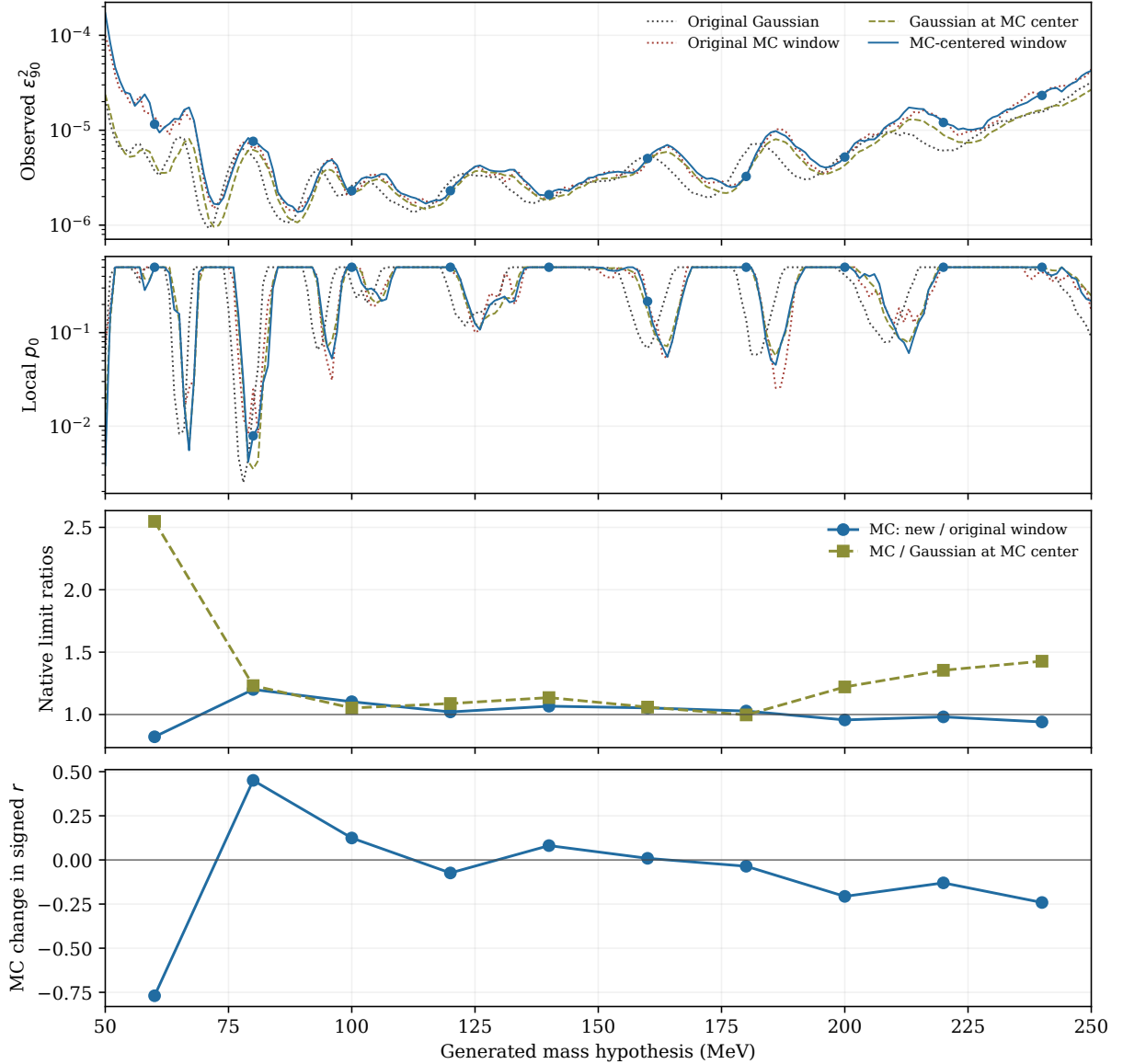


Figure 9: Original and MC-centered observed inference for the 2021 10% sample. Native MC points are marked; continuous MC curves include exploratory morphs. The bottom panels compare only the ten generated masses. The matched Gaussian comparison separates the remaining shape difference from placing a Gaussian peak at the wrong reconstructed position.

12.2 Native-mass limits and local probabilities

m_0	Old MC limit	New MC limit	New/old	Old MC p_0	New MC p_0	New G p_0
60	1.41×10^{-5}	1.16×10^{-5}	0.822	0.49502	0.50000	0.50000
80	6.36×10^{-6}	7.64×10^{-6}	1.201	0.02477	0.00787	0.00348
100	2.09×10^{-6}	2.30×10^{-6}	1.103	0.50000	0.50000	0.50000
120	2.26×10^{-6}	2.30×10^{-6}	1.021	0.50000	0.50000	0.50000
140	1.96×10^{-6}	2.09×10^{-6}	1.067	0.50000	0.50000	0.50000
160	4.79×10^{-6}	5.04×10^{-6}	1.053	0.21877	0.21609	0.17038
180	3.17×10^{-6}	3.26×10^{-6}	1.028	0.50000	0.50000	0.50000
200	5.45×10^{-6}	5.22×10^{-6}	0.957	0.50000	0.50000	0.50000
220	1.23×10^{-5}	1.21×10^{-5}	0.981	0.50000	0.50000	0.50000
240	2.47×10^{-5}	2.32×10^{-5}	0.940	0.44359	0.50000	0.50000

Table 11: Original and shifted-window MC results. Limits use the inherited coupling coordinate. The last column is the Gaussian benchmark centered at the same MC core. All probabilities are local asymptotic references.

m_0	Gaussian at c limit	MC/G at c	$r_{MC} - r_G$	Old MC fit (%)	New MC fit (%)
60	4.54×10^{-6}	2.548	-0.520	39.26	39.65
80	6.20×10^{-6}	1.232	-0.284	67.56	70.95
100	2.19×10^{-6}	1.052	-0.292	73.19	78.21
120	2.12×10^{-6}	1.088	-0.076	74.60	80.37
140	1.84×10^{-6}	1.136	+0.068	76.52	81.30
160	4.76×10^{-6}	1.059	-0.167	75.22	81.75
180	3.27×10^{-6}	0.997	-0.434	75.27	80.46
200	4.27×10^{-6}	1.221	-0.002	75.71	79.37
220	8.91×10^{-6}	1.355	+0.073	73.86	77.30
240	1.63×10^{-5}	1.427	-0.048	72.38	74.97

Table 12: Matched centered-window comparison and retained MC probability in the fitted bins. Support normalization remains unchanged; shifting the mask does not renormalize its retained template fraction to one.

At the native 80 MeV hypothesis, the reconstructed center is 78.10 MeV. The shifted MC upper limit is 7.639×10^{-6} , compared with 6.362×10^{-6} before shifting, a 20.08% increase. The local probability decreases from 0.02477 to 0.007875; an increased upper limit and a smaller local probability are compatible because they answer different likelihood questions.

On the dense exploratory MC grid the smallest local probability moves to 50 MeV ($p_0 = 0.003798$). That template uses the previously failing low-mass interpolation region. It is not promoted to a validated search result; the native comparison above remains the primary result. The unchanged selected-candidate association and v13/v16 production qualifications still apply. No global probability, coverage calibration or physically validated A-prime exclusion follows from this recentering.

12.3 Reconstructed cores and window placement

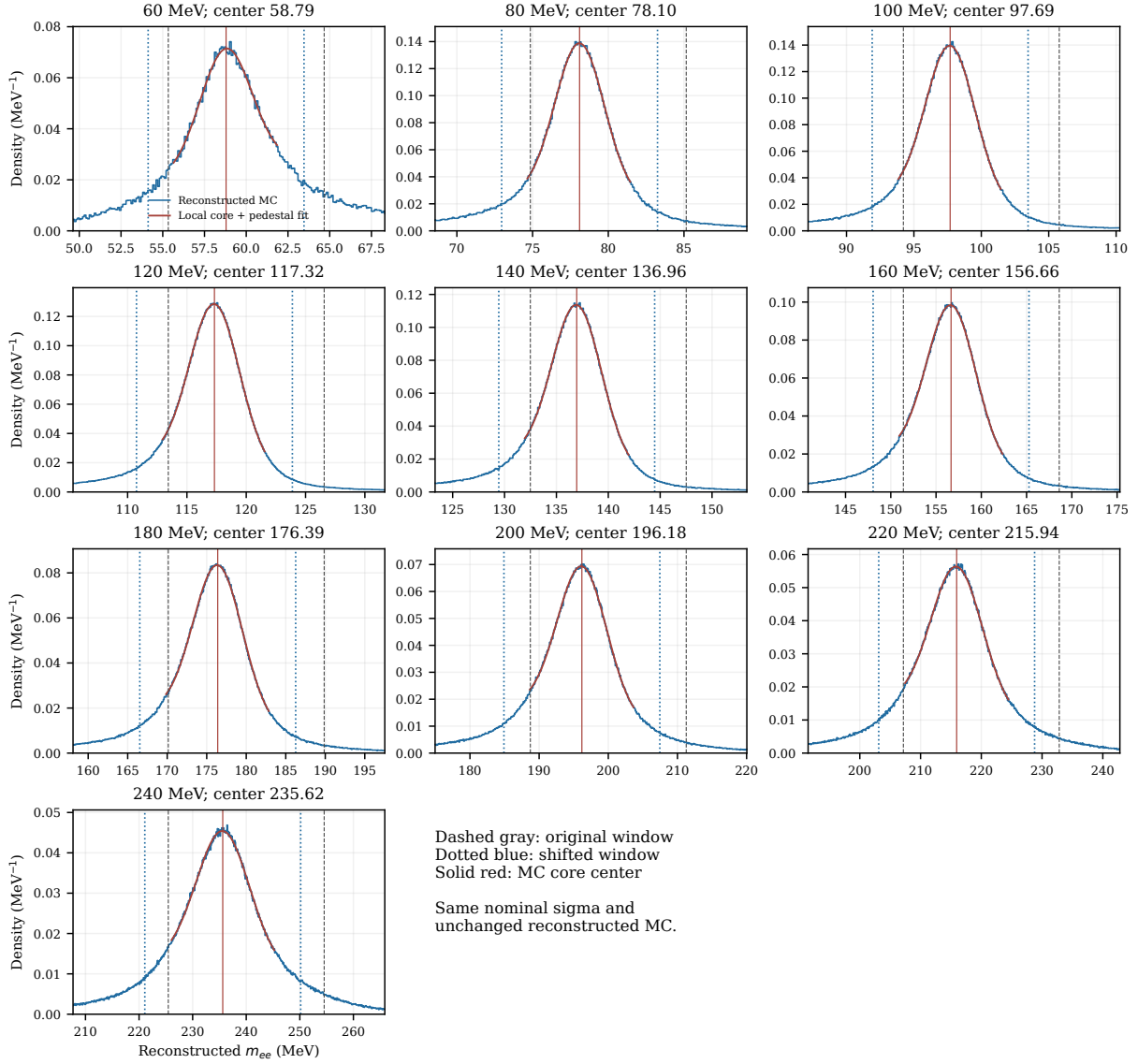


Figure 10: Native selected-MC cores and the original versus shifted $\pm 2.25\sigma_{\text{nominal}}$ windows. Blue curves are unchanged empirical templates. The local Gaussian plus affine-pedestal fit (red) estimates the center only and is never used to replace the full MC signal shape in the observed likelihood. The red vertical line marks the fitted core center.

13 An analytic description of the MC core shift

The ten native cores within the 60–240 MeV study range support a compact empirical location law. Four simple forms are fitted with equal weight per mass: a proportional shift, an affine function, a quadratic and a logarithm. Each mass is also omitted in turn and predicted from the other nine. No observed spectrum enters this comparison. The best leave-one-mass-out (LOO) RMS among these candidates is the logarithmic model,

$$c(m) - m = -3.22433 - 2.21399 \ln\left(\frac{m}{150}\right) \text{ MeV}, \quad 60 \leq m \leq 240 \text{ MeV}. \quad (8)$$

The argument uses m in MeV. The law is an empirical approximation, not a detector mechanism or an extrapolation outside the fitted range. Its fitted RMS is 0.0497 MeV, LOO RMS 0.0652 MeV and largest held-out error 0.1604 MeV. These errors describe cross-mass prediction, not a calibrated confidence band. Model selection and evaluation use the same ten masses; the LOO summary is a diagnostic, not independent validation.

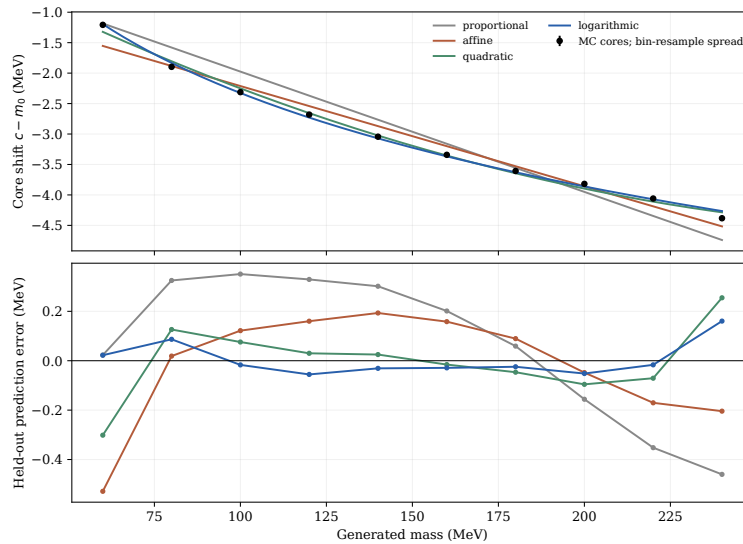


Figure 11: Measured core shifts, analytic models and their held-out residuals. Small error bars show conditional MC-bin resampling spread; the equal-mass fits do not treat that spread as the complete response uncertainty.

Model	Parameters	Fit RMS (MeV)	LOO RMS (MeV)	Max. LOO (MeV)
proportional	1	0.2552	0.2885	0.4602
affine	2	0.1555	0.2154	0.5287
quadratic	3	0.0679	0.1399	0.3015
logarithmic	2	0.0497	0.0652	0.1604

Table 13: Analytic model comparison. Affine and quadratic forms use $x = (m - 150)/100$; coefficients and all held-out predictions are saved.

14 Can one normalized core describe all masses?

The ten distributions are first recentered without rescaling their physical widths. They are then expressed in $u = (m_{ee} - c)/\sigma_{\text{core}}$ using the measured local Gaussian width. The density transformation includes the Jacobian: $q_m(u) = \sigma_{\text{core}} p_m(c + \sigma_{\text{core}} u)$. Each full source histogram has unit area over its recorded 0–400 MeV range. Display cropping never renormalizes the full distributions. The separately labeled core view conditions on $|u| < 2$ and therefore discards information about the fraction of all candidates in that core.

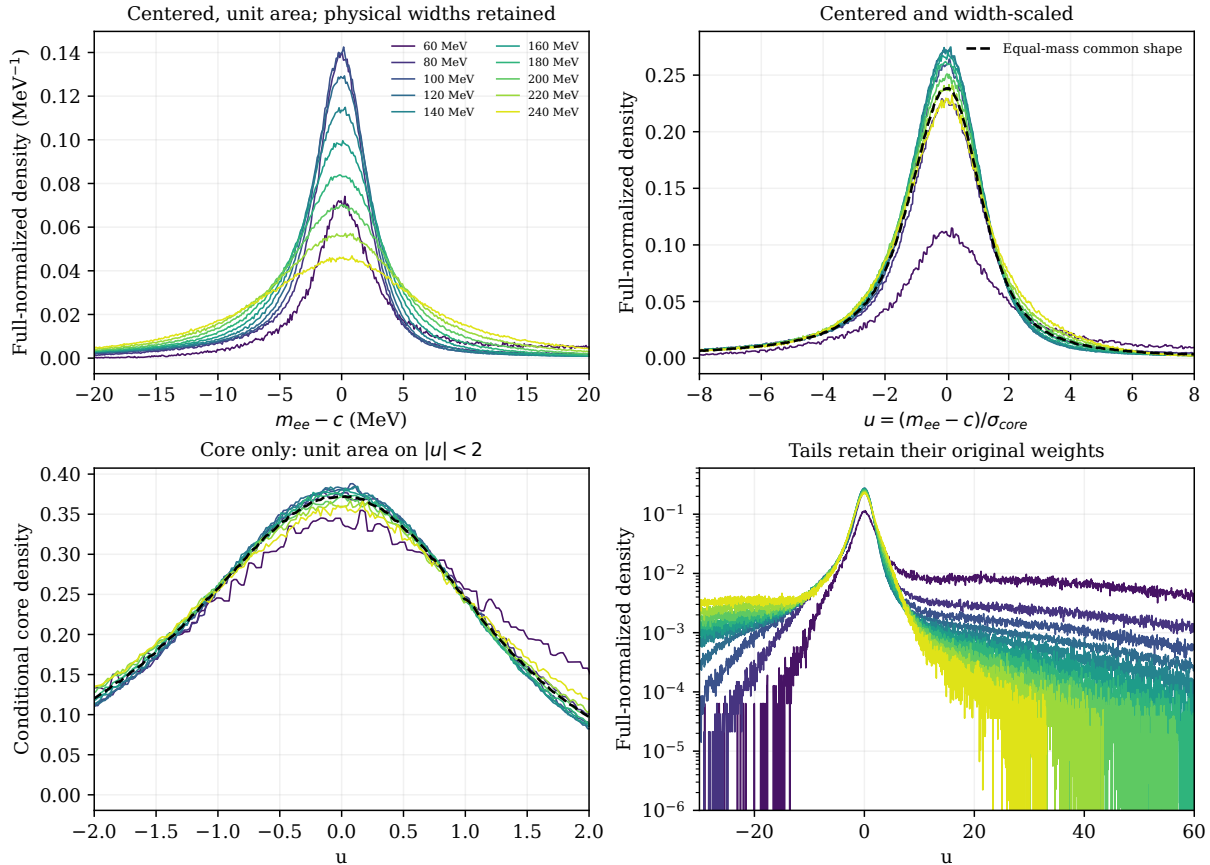


Figure 12: All ten normalized native MC histograms superimposed; colored curves are empirical histograms, not Gaussian fits or morphed samples. Top left: centered physical-mass densities. Top right: full-normalized densities after width scaling. Bottom left: unit-area conditional cores. Bottom right: full-normalized tails on a log scale. The black dashed curve is the equal-mass mean standardized distribution; sample event counts do not determine its weights. Direct MC-point overlays and the rebinning definition appear in Section 16.

The conditional cores are much closer than the full shapes. A shared central shape is therefore a useful possibility, but its unit-area overlay cannot establish a common core fraction or common tails. The full candidate distributions include the same unresolved response and candidate-association contributions as the earlier sections.

14.1 Held-out tests of a common shape

For each generated mass, the common standardized CDF is averaged from the other nine histograms. The primary shape-only test supplies the held-out sample’s known center and width. A second test predicts its center from the selected analytic law fitted to the other nine and predicts its width from a quadratic in $\ln(\sigma_{\text{core}}/\text{MeV})$ versus $x = (m - 150)/100$, also fitted without that sample. All likelihood comparisons use the same measured-center evaluation window and GP constraint as the direct MC. The second test predicts the complete template but does not also move its evaluation mask. Existing neighboring-mass morphs are evaluated in that same window as a comparison.

m_0	Core (%)	D_{core}	D_{full}	Shared UL ratio	Morph UL ratio	Predictive UL ratio
60	32.4	0.045	0.461	0.443	1.047	0.419
80	60.7	0.009	0.075	0.944	1.324	0.991
100	68.1	0.008	0.023	1.140	1.050	1.155
120	71.0	0.010	0.061	1.191	1.025	1.172
140	71.5	0.010	0.072	1.135	1.009	1.124
160	70.7	0.009	0.067	1.135	1.016	1.127
180	69.6	0.007	0.052	1.085	1.017	1.076
200	67.5	0.007	0.033	1.002	1.003	1.003
220	65.8	0.009	0.037	0.936	1.018	0.936
240	63.6	0.014	0.049	0.864	1.005	0.860

Table 14: Held-out CDF distances and observed upper-limit ratios relative to direct MC. Core probability is conditional on the stored histogram range; D_{core} conditions on $|u| < 2$. Full distances use the declared analysis support. “Predictive” estimates center and width as well as shape without the held-out histogram.

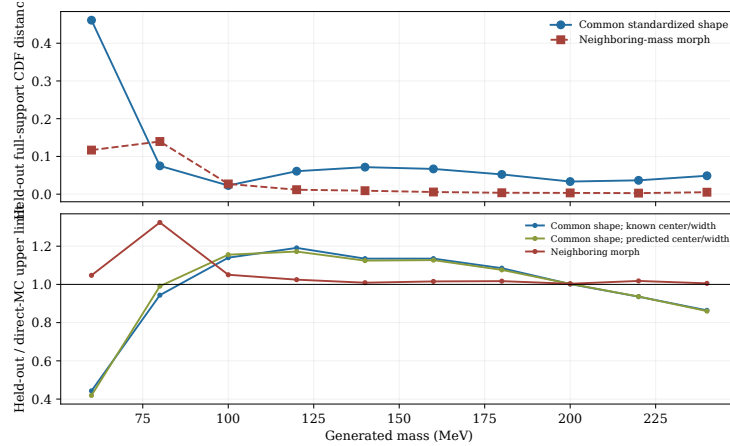


Figure 13: Common-shape and neighbor-morph prediction errors, with matched likelihood conditions.

Core-conditioned CDF distances span 0.0067–0.0450, while full-support distances span 0.0229–0.4611. The common-shape limit ratios span 0.443–1.191 even when location and width are supplied. Thus a single full standardized shape is not supported as a drop-in replacement. A shared core with mass-dependent core fraction and tails is better motivated by these diagnostics. For inspection, 181 exploratory common-shape templates are saved at 1 MeV spacing over 60–240 MeV using the analytic center and width laws; they are not adopted as the nominal signal model.

15 Left and right signal probability and asymmetric blinding

Asymmetry is measured around the fitted core, retaining the full stored-histogram normalization. The upper panel measures probability beyond $\pm 2.25\sigma_{\text{nominal}}$; the lower panel compares the two halves inside that interval. Source overflow is below 0.016% and is excluded by the explicitly declared 0–400 MeV normalization here.

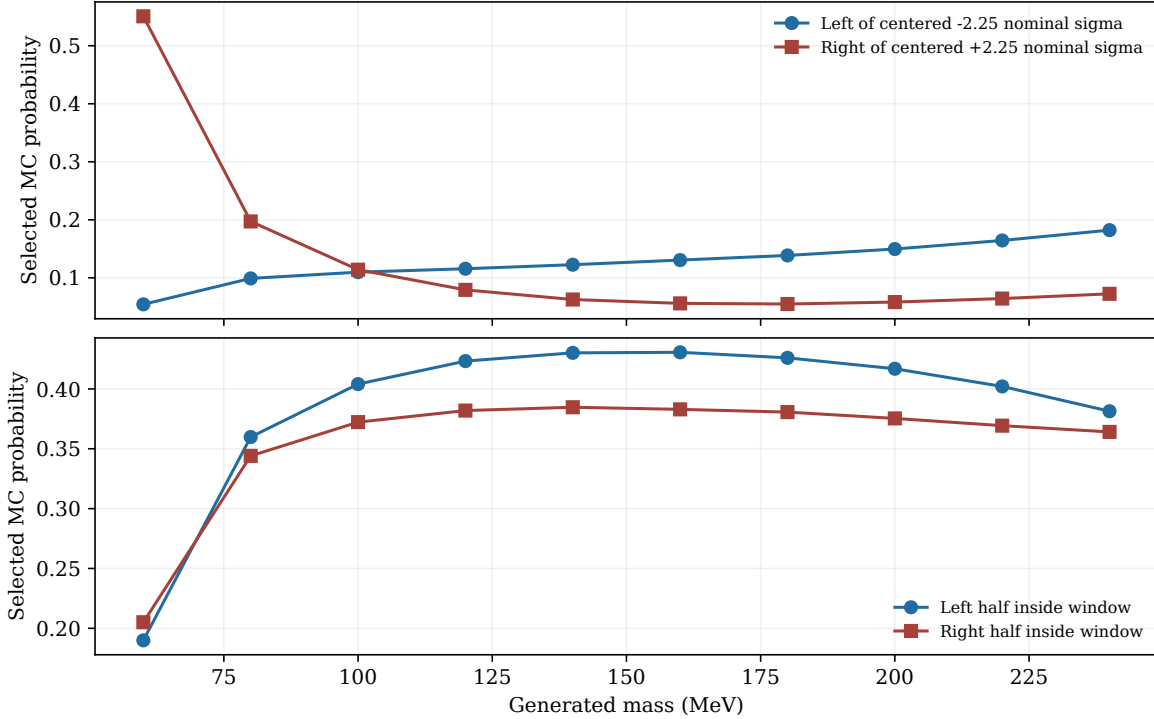


Figure 14: Mass dependence of the left and right selected-MC contributions around the core.

The left external tail is larger at 120–240 MeV, whereas the right external tail is larger at 60–100 MeV. The direction is therefore not universal across the accepted samples.

Five windows are declared around the same measured core before fitting the observed data: $[-2.25, +2.25]$, $[-2.5, +2.0]$, $[-2.0, +2.5]$, $[-2.75, +2.25]$ and $[-2.25, +2.75]$, all in nominal sigma. The redistributed pair keeps total width 4.5 sigma; each is also exactly a shift of the window midpoint by ∓ 0.25 sigma, so it cannot isolate tail asymmetry from center placement. The expanded pair adds half a sigma on either side as matched controls.

Each window defines both the fitted bins and excluded GP training bins. The GP prediction and covariance are recomputed; archived kernel, resolution and coupling conventions remain anchored at the generated mass. Gaussian and direct-MC templates share the same mask and background. Actual bin counts are saved because equal continuous widths need not select the same number of discrete observed bins.

15.1 Observed asymmetric-window comparisons

Window in nominal sigma	MC UL / symmetric	Med. $ MC/G - 1 $ (%)	Min. mass	Min. local p0
$[-2.25, +2.25]$	1.000–1.000	17.8	80	0.00787
$[-2.5, +2]$	0.942–1.038	19.4	80	0.00671
$[-2, +2.5]$	0.964–1.119	18.6	80	0.00854
$[-2.75, +2.25]$	0.983–1.129	20.5	80	0.00761
$[-2.25, +2.75]$	0.997–1.133	18.5	80	0.00868

Table 15: Prespecified native-mass window comparisons. The minimum local probability is descriptive and is not corrected for testing several windows. No prescription is chosen from the observed result.

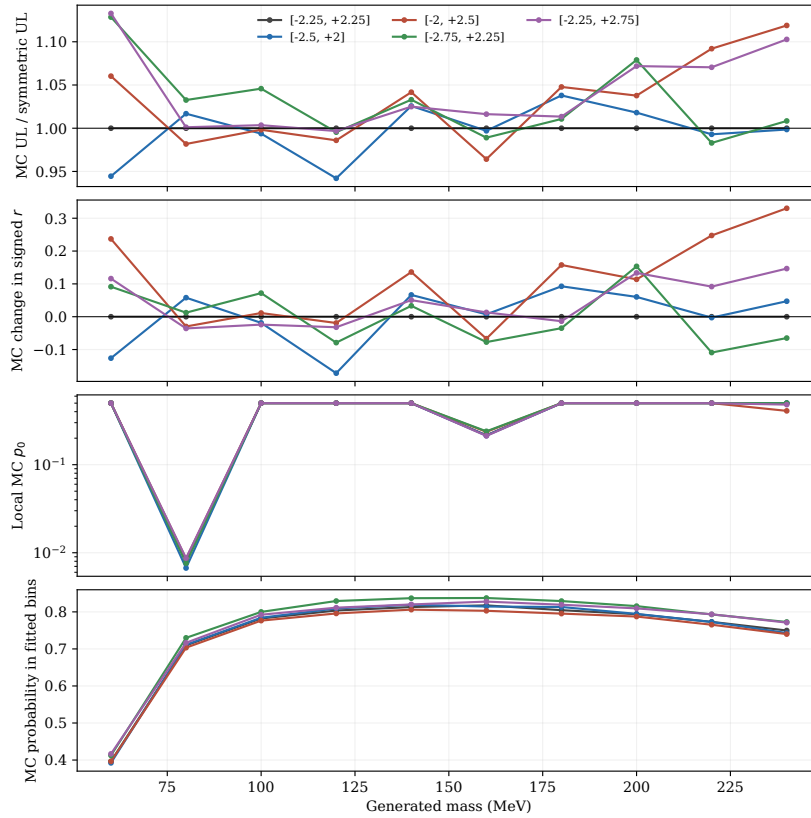


Figure 15: Native MC limit ratios, signed-root changes, local probabilities and fitted signal fractions for all five windows. Full MC/Gaussian limits and probabilities are retained in the accompanying 100-row fit ledger. Changes include altered data bins and altered background interpolation, as well as signal acceptance.

These are conditional observed diagnostics. A preferred production window would require signal-recovery and coverage checks, including signal leakage into the training sidebands, rather than selection by the largest observed local significance or tightest upper limit.

16 What the Figure 12 curves represent

The colored curves are the ten native reconstructed-MC histograms, normalized and plotted as connected bin densities. The local Gaussian plus affine-pedestal fit supplies only c and σ_{core} ; it does not supply their line shapes. In the standardized panels, a piecewise-linear CDF through the native 0.1 MeV histogram edges is integrated into bins of width $\Delta u = 0.02$. This is empirical rebinning, with uniform density within each original bin, rather than an analytic signal parameterization. The dashed black curve is a separate candidate: the equal-mass mean of these standardized shapes.

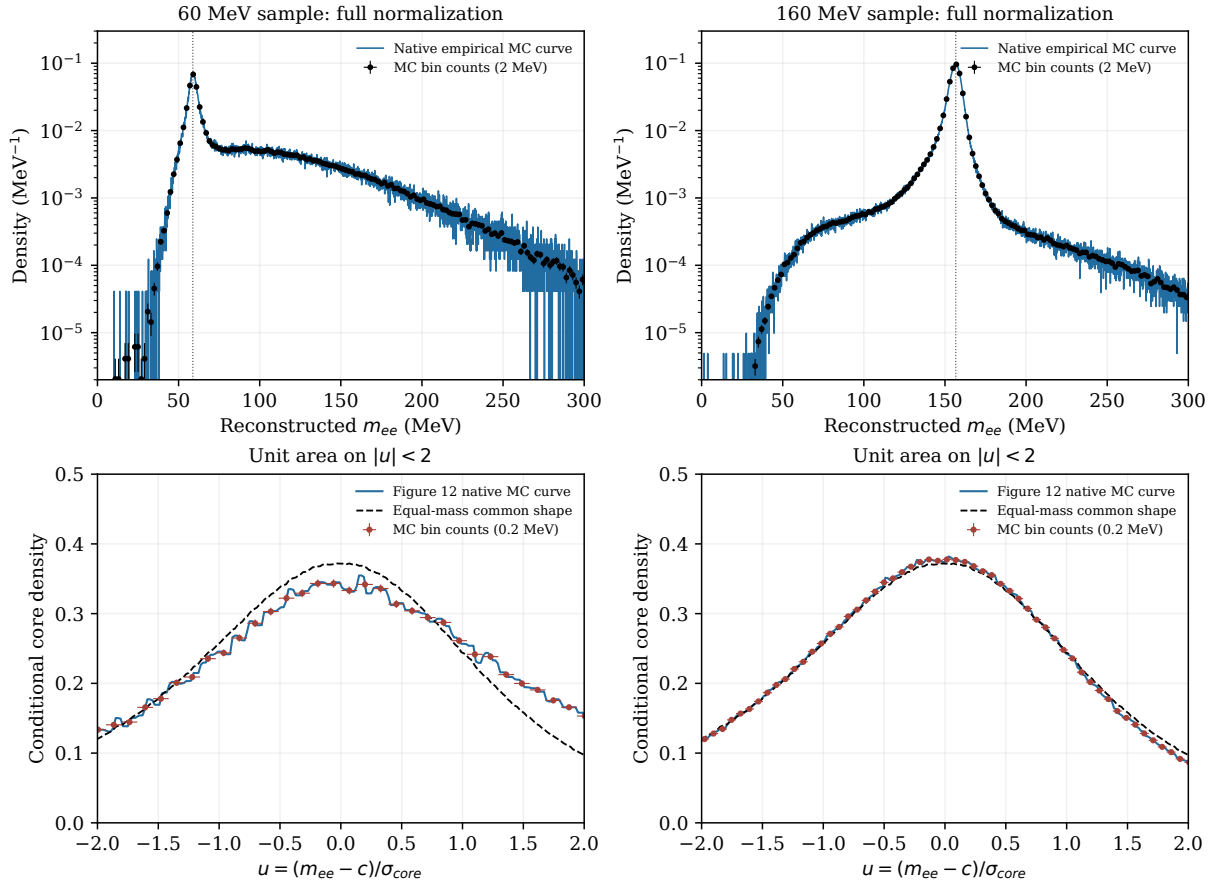


Figure 16: Direct MC-point overlays for the outlying 60 MeV sample and a representative 160 MeV sample. Top: full-normalized native empirical curves and counts summed into 2 MeV bins. Bottom: exactly the standardized empirical curves of Figure 12 and direct sums into 0.2 MeV bins, both conditioned on $|u| < 2$. Horizontal bars show bin widths; vertical bars show $\sqrt{\sum w^2}$ with the display normalization held fixed. The dashed common shape is shown separately. Points are bin averages; the curve must be integrated over the same bins for an exact comparison.

The native curves agree with the MC points by construction: the largest difference between a direct bin sum and the corresponding CDF integral is below 10^{-9} candidates. This verifies the plotted representation, not an independent goodness of fit; no fit probability is assigned to that identity. The common-shape candidate need not agree with each sample, and the held-out tests above assess that separate question. The plot and all underlying counts are saved.

17 The 60 MeV outlier and the meaning of a window in u

At 60 MeV, 32.4% of the selected histogram is inside $|u| < 2$, compared with 60.7–71.5% at the other nine masses. Around the current centered $\pm 2.25\sigma_{\text{nominal}}$ window, 5.4% is on the left, 55.1% on the right and 39.5% inside. Thus the full-normalized core is small because much of this selected-candidate distribution lies elsewhere. The core-conditioned panel removes that fraction information and still shows some shape variation.

The 60 MeV files contain 243,668 selected rows. Their event cutflows record 1,542,086 readout events and the same 243,668 events at the final selection stage, a conditional retention of 15.8%. Other masses give 34.5–61.6% (Table 16). This supports lower retention from the recorded readout stage, but neither that stage nor the unit-normalized Figure 12 establishes generated-signal acceptance. A certified generated-event denominator and signal-daughter association are still needed; the selected candidate distribution is not certified as a pure signal response. “Within the study range” does not mean high detector acceptance.

The existing core-centered window and a proposed fixed- u window are different because u uses the fitted core width:

$$|m_{ee} - c| < 2.25\sigma_{\text{nominal}} \iff |u| < 2.25 \frac{\sigma_{\text{nominal}}}{\sigma_{\text{core}}} = 2.920\text{--}3.174. \quad (9)$$

Consequently $|u| < 2.25$ would narrow the physical window by 23.0–29.1%. At 60 MeV, $c = 58.792$ MeV, $\sigma_{\text{core}} = 1.550$ MeV and $\sigma_{\text{nominal}} = 2.075$ MeV: the half-width changes from 4.669 to 3.488 MeV and the retained selected-MC fraction falls from 39.5% to 34.5%.

m_0	σ_{core}	σ_{nom}	Old half-width (u)	Old MC (%)	$\pm 2.25u$ MC (%)	Selected/readout (%)
60	1.550	2.075	3.012	39.5	34.5	15.8
80	1.637	2.288	3.145	70.4	63.6	34.5
100	1.853	2.570	3.120	77.6	71.1	47.5
120	2.112	2.920	3.111	80.5	74.0	55.3
140	2.392	3.339	3.140	81.5	74.6	59.5
160	2.713	3.827	3.174	81.3	73.9	61.3
180	3.120	4.383	3.161	80.7	72.9	61.6
200	3.577	5.008	3.150	79.2	71.0	60.0
220	4.233	5.702	3.031	77.1	69.4	57.0
240	4.981	6.465	2.920	74.5	67.3	52.8

Table 16: Widths in MeV and continuous-window probabilities from the stored 0–400 MeV MC histograms. “Old” means the existing core-centered $\pm 2.25\sigma_{\text{nominal}}$ window. The last column is conditional readout-stage retention, not detector acceptance. Actual observed-bin masks use bin centers and can differ slightly from these continuous fractions.

A fixed- u prescription is a coherent MC-based candidate, but the coefficient must be declared as a new choice. It is not a change of units that preserves the old blind region. Nor does a non-Gaussian empirical shape inherit a Gaussian containment fraction from its fitted core width. This clarification does not change the primary mask or recompute observed limits and local probabilities. Testing a new mask would require the matching GP training exclusion and signal-recovery checks, as for the preceding width diagnostics.